Signal detection methodology in spontaneous reporting systems: a narrative review

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Abstract

This narrative review surveys statistical approaches to signal detection in spontaneous reporting databases. We compare proportional reporting ratios, reporting odds ratios and Bayesian shrinkage estimators, and discuss the effect of reporting biases on each. No individual patient data are presented and no new cases are described.

Keywords: signal detection, disproportionality, methodology, review

Introduction

Spontaneous reporting systems collect large volumes of unstructured, self-selected reports. Turning that into a reliable signal is a statistical problem as much as a clinical one. This review is intended for readers who assess signals operationally rather than for methodologists.

Discussion

Every measure discussed shares a dependence on the denominator problem: the number of patients exposed is not known. Bayesian shrinkage mitigates the instability of ratios computed on small counts but cannot correct a biased numerator. Stimulated reporting after media coverage remains the single largest practical confounder.

Conclusion

No single statistic is sufficient on its own. Quantitative screening should prioritise clinical review, not replace it.

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Conflict of interest

The authors declare that they have no competing interests.

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